

Quantity	At Perihelion	At Aphelion
Main accel. GM/r^2 (m/s ²)	6.272×10^{-2}	2.722×10^{-2}
J2 perturbation (m/s ²)	9.476×10^{-12}	1.785×10^{-12}
Ratio J2/main ($\times 10^{-7}$)	1.511×10^{-10}	6.558×10^{-11}